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Proposal: Forced Alignment of Speech into MIDI

I find myself drawn to the effect that non-lexical factors, like cadence and volume, have on the meaning of spoken language audio data. I propose a project that will train a model to perform forced alignment to find the specific timestamps for the beginning and end of each word in a continuous spoken sequence, given a transcript. Some additional engineering will then be applied to format the discrete collection of words as a MIDI file, allowing the use of traditional audio engineering and music theory analysis techniques to look at things like pitch, tempo, cadence/rhythm and volume. My goal is to expose these properties for use in a handful of different explorations and downstream applications.

How closely can I predict the boundaries of different words given a set of labeled data like TIMIT? What factors confuse the model's ability to predict word boundaries? How much information can be effectively represented as MIDI data, and how much is lost in that conversion? I hypothesize that the biggest impacts will come from: variation in pronunciation of the same words, the rate of speech and the amount of pause between phonetic sounds.

Other research in this area seems to approach boundary detection using the Viterbi algorithm, and Hidden Markov Models or applies CTC on neural networks. The problems that have arisen in works using HMMs and Viterbi are: the unrealistic duration of phonemes that arise from the HMM model and some misaligned word boundaries. Based on all of this previous work, and the fact that we have worked with Viterbi and HMM's in class, this is what I will use to train my model as well.

As an aside, the downstream applications of using the MIDI file for analysis are also interesting and serve as motivation. Is an analysis of volume changes (dynamics), pitch changes (melody), natural pauses and spaces between word sequences (rhythm), and changes in rate of speech (tempo) useful in the field of natural language processing? How would such information impact sentiment analysis, subtext collection, or speech classification tasks? It's also worth noting that this project has some rather fun applications in traditional audio-engineering contexts.

Citations

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